Paritosh Singh

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Education

SRM Institute of Science and Technology, Chennai Jun 2023 to May 2027
B.Tech, Computer Science and Engineering (Data Science) *CGPA 8.78/10; Class XII 76.6%; Class X 84.4%*
Relevant Coursework: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

Software Development Engineer in Test Intern Jun 2026 to Present
HealthEdge *Bengaluru, India*

- Roughly doubled the failing tests the team triages per day with a Python tool that parses failed Katalon runs, sorts each into one of 15 failure signatures across 7 categories and drafts a fix.
- Built zephyr-migrate, an internal TypeScript MCP server that lets coding agents turn Zephyr test cases into Katalon tests; a static gate rejects any method, locator or value not already in the repo.
- Grounded every drafted fix in retrieval (RAG) over the repo's locators, custom keywords and past failure logs instead of the model's memory, and reran each fix headless through the Katalon CLI before it reached code review.
- Merged 150+ Groovy UI regression scripts for a healthcare payer web app through code review in two week sprints.

Projects

mini-redis | *C++, POSIX sockets, epoll, Linux* github.com/Paritoshcodes/mini-redis

- Served 150K+ SET and GET requests per second from 50 concurrent clients on one core at about 0.2 ms p50 latency, using a single threaded, non blocking epoll loop over raw POSIX sockets, so the hot path needs no locks.
- Pushed GET throughput to 2.7M+ per second at 128 deep pipelining by making the RESP parser buffer partial frames, so a command split across two TCP reads is parsed once it is complete instead of dropped.
- Implemented 18 commands in 594 lines of C++, including SET with EX/PX, TTL, INCR and pub/sub with lazy expiry checked on access; 1,000 of 1,000 pub/sub messages delivered under test, driven by stock redis-cli.

TeamPulse | *Node.js, Express, MongoDB, Redis, Socket.IO, WebRTC, React* team-pulse-client-x2oh.vercel.app

- Wrote the Express and MongoDB backend for a Slack style collaboration app: 73 REST endpoints and 15 Socket.IO event handlers covering channels, direct messages, typing presence and WebRTC call signalling.
- Traced video calls that never connected to sockets that were not joining their per user rooms, so call signals went to empty rooms; routing signalling through per user Socket.IO rooms fixed it.
- Restored signup OTP email after Railway blocked outbound SMTP by switching delivery to Resend's HTTP API.

Revenant | *Python, FastAPI, TypeScript, PostgreSQL, Playwright* github.com/Paritoshcodes/revenant

- Built a Razorpay AI Buildathon agent that diagnoses failed payments and retries them through a Playwright checkout.
- Kept the LLM out of money decisions: it only names a broad failure class and deterministic code picks the retry, which closed a prompt injection path that schema validation could not catch.
- Measured causal lift, not gross recovery, with a stratified 50/50 holdout; against a frozen true lift over 2,000 runs the 95% interval covered it 93.8% of the time, reported as slightly anti-conservative rather than tuned.

Vigil | *Python, FastAPI, PostgreSQL, XGBoost, LightGBM, SHAP, React* github.com/Paritoshcodes/Vigil

- Predicted seven natural hazards for 25 Indian cities at ROC AUC 0.74 to 0.88 by hazard (heatwave 0.876, cyclone 0.852) with XGBoost and LightGBM ensembles, served through FastAPI and a React dashboard.
- Added seasonal bias correction, adaptive thresholds and confidence weighted fusion; method published in ISJEM.

Agent Replay | *Python, Strands Agents, AWS Bedrock, Lambda, DynamoDB* github.com/Paritoshcodes/agent-replay

- Built a CI gate for AI agents at the WeMakeDevs x AWS hackathon that records 5 runs, derives a YAML contract of required calls, allowed variation and data dependent order, and fails the PR at the first divergent step.
- Froze tool results from the recordings while keeping the model live, so each test changes exactly one variable; arguments that trace to nothing the model saw are flagged as fabricated, and 339 offline tests fail if code opens a socket.
- Deployed the opt-in backend as 4 independent CloudFormation stacks (DynamoDB run index, content-addressed S3, Lambda read and fork APIs), with CI assuming a repo-scoped IAM role via OIDC so no AWS keys exist anywhere.

Open Source Contributions

- **MapLibre Martin** (Rust, 3.9K stars): fixed a .pbf redirect ignoring `--route-prefix`; merged with a regression test as PR #2599 and shipped in release v1.10.1.
- **render-engine-cli** (Python): fixed sitemap URLs when serving locally; merged in PR #40 after two review rounds.

Technical Skills

Languages: Java, C++, Python, SQL, TypeScript, JavaScript, Groovy
Frameworks: Node.js, Express, FastAPI, React, Socket.IO, Strands Agents, pytest, Jest, Playwright
Databases: PostgreSQL, MongoDB, Redis, DynamoDB
Developer Tools: Git, GitHub Actions, Docker, Linux, AWS (Lambda, S3, Bedrock, CloudFormation), Postman, Katalon
Libraries: pandas, NumPy, scikit-learn, XGBoost, LightGBM, SHAP

Certifications and Achievements

Certifications: SAP Certified Back End Developer, ABAP Cloud (2026); NPTEL Machine Learning, IIT Madras, Elite (2024)
Achievements: 2nd place, Capture the Flag, SRMIST (2024); Associate Corporate Lead, SRMKZILLA (2024 to 2025)